

The Iapetus's ridge: Possible explanations of its origin

L. Czechowski^{a,*}, J. Leliwa-Kopystyński^{a,b}

^a *Institute of Geophysics, University of Warsaw, Pasteura 7, 02-093 Warsaw, Poland*

^b *Space Research Center of Polish Academy of Sciences, Bartycka 18A, 00-716 Warsaw, Poland*

Received 3 November 2006; received in revised form 18 July 2007; accepted 8 August 2007

Abstract

Images from the Cassini mission show the existence of a long ridge on Iapetus. It extends at least 1400 km along the great circle defined by the equator. We discuss the possibility that the ridge is a result of extensional forces acting above an ascending current of solid-state convection. A two-cell pattern of convection is a reasonable explanation of the observed feature. Three scenarios of the ridge formation are proposed: spin–orbit resonance scenario, convection in low viscosity-interior scenario, and impact generating flow scenario.

© 2007 COSPAR. Published by Elsevier Ltd. All rights reserved.

Keywords: Geological processes; Iapetus ridge; Satellites of Saturn

1. Introduction

Iapetus, a medium size icy satellite (MIS), is considered in this paper. The lower limit of MIS radii is approximately 200 km, which corresponds to the smallest spherical icy bodies (e.g. Czechowski, 2006). The upper limit, some 800 km, corresponds to the bodies where water ice is mostly in the form of low-pressure polymorph (ice I). Discussion of the stratification of water ice phases inside the icy satellites is given by Leliwa-Kopystyński et al. (1994). Discussion concerns both for non-differentiated and differentiated models of the satellites interiors. Fourteen bodies (6 satellites of Saturn, 5 of Uranus, 2 of Neptune, and 1 of Pluto) belong to the MIS class.

The authors of this paper considered a possibility of internal convection within the MIS in a few previous papers (e.g. Czechowski and Leliwa-Kopystyński, 2002a,b, 2003, 2005 later on cited as Paper I). The new data provided by the Cassini mission in January 2005 prompted

us to consider the possible scenarios of the evolution of Iapetus.

According to Thomas et al. (2006) the physical data of Iapetus are: mean radius $d = (736.0 \pm 2.0)$ km and mean density $\rho = (1081 \pm 18)$ kg m⁻³. The axes of Iapetus' ellipsoid are $a = 747.1$ km (the Saturn-facing), $b = 749.0$ km (the orbit-facing) and $c = 712.6$ km (polar). Note two unusual facts (i) the longest dimension is parallel to the orbit but it is not Saturn-facing; (ii) the shape of Iapetus corresponds to a spin period of ~ 17 h, instead of the present value 79 days. We believe that the first fact indicates for the large-scale non-homogeneities of density.

Some physical data, as well as the mean orbital elements referred to the local Laplace plane (according to http://ssd.jpl.nasa.gov/sat_elem.html, updated November 30, 2005)¹ are given in Table 1. The relatively large size of Iapetus makes it unlikely that its interior has retained any porosity that it may have inherited from the time of its formation. The evolution of porosity in the icy bodies is discussed by Leliwa-Kopystyński and Kossacki (2000);

* Corresponding author. Tel.: +48 22 5546850; fax: +48 22 5546882.
E-mail addresses: lczech@fuw.edu.pl (L. Czechowski), jkopyst@mimuw.edu.pl (J. Leliwa-Kopystyński).

¹ http://ssd.jpl.nasa.gov/sat_elem.html; http://ssd.jpl.nasa.gov/?sat_phys_par; <http://saturn.jpl.nasa.gov/multimedia/images/image-details.cfm?imageID=1270>.

Table 1
Comparison of data concerning three Saturnian satellites

Parameter (reference)	Tethys	Rhea	Iapetus
d radius [km] (1)	536.3 ± 1.5	764.5 ± 2.0	736.0 ± 2.0
Gm [$10^9 \text{ m}^3 \text{ s}^{-2}$] (1)	41.210 ± 0.007	154.07 ± 0.16	120.50 ± 0.03
ρ density [kg m^{-3}] (1)	956 ± 8	1233 ± 10	1081 ± 18
P orbital period [days] (2)	1.888	4.518	79.33
$n = 2\pi/P$ orbital mean motion [rad s^{-1}]	$3.852 \text{E}-05$	$1.610 \text{E}-05$	$9.167 \text{E}-07$
e eccentricity (2)	0.0001	0.0009	0.0284
r semimajor axis [km]	294 660	527 040	3 561 300
$\psi = Mr^3/(ma^3)$ tidal parameter	$5.55 \text{E}-03$	$7.513 \text{E}-04$	$2.763 \text{E}-06$
$\chi = \rho n^5 d^4 e^2$, Eq. (4)	$6.7 \text{E}-05$	$3.7 \text{E}-04$	$1.65 \text{E}-07$
q_{rad} radiogenic heating [W kg^{-1}]	$1.32 \text{E}-13$	$1.87 \text{E}-12$	$1.07 \text{E}-12$
$q_{\text{tidal}}/q_{\text{tidal_Rhea}}$	0.18	1	$0.45 \text{E}-03$
$\tau = mr^6 \omega_p / (GM^2 d^3 \delta)$ is despinning time [yr] in non-resonant regime. Assumed: primordial spin period is 10 h and $\delta = 0.01$ (3)	$7.1 \text{E} + 01$	$2.9 \text{E} + 03$	$2.5 \text{E} + 08$
Total length of long linear tectonic features [km]	2400 (Paper I)	0 (Paper I)	1400 (4)
Tectonic activity parameter ζ	2.04	0	0.58

Radiogenic heating is calculated using the following values: density of water ice at 100 K is 940 kg m^{-3} , density of mineral component 3060 kg m^{-3} , rate of the radiogenic heating (following Schubert et al., 1986) in chondritic material $5.45 \times 10^{-12} \text{ W kg}^{-1}$. The tectonic activity parameter ζ is a simple quantitative measure of such activity: $\zeta = (\text{observed total length of the long linear structures of tectonic origin})^2 / [4\pi d^2 \times (\text{fraction of the mapped surface})]$; see Paper I for more details.

References: 1 – http://ssd.jpl.nasa.gov/sat_phys_par; 2 – http://ssd.jpl.nasa.gov/sat_elem.html; 3 – Hubbard (1984); 4 – Porco et al. (2005); Radius and density of Iapetus according to Thomas et al. (2006).

see also the references therein. Low density of Iapetus shows that it is mainly composed of ice and that its silicates content is about 22% by mass or about 8% by volume. The predominant ice-forming phase is likely to be water (see e.g. Roush, 2001). However, the presence of admixtures of other volatiles is possible (e.g. ammonia and methane, see Hunten et al., 1984; Prentice, 2005). Any of them would make the viscosity of the interior lower (see e.g. Durham et al., 1993) and therefore it makes the development of convection easier.

After Voyagers 1 and 2 encounters with the Saturnian System Iapetus was considered to probably be a geologically dead body. Schubert et al. (1986) state that the solid-state convection in Iapetus lasted probably for a significant period of time. However, the most of Iapetus thermal history was dominated by conductive cooling. The most intriguing feature of Iapetus observed during the Voyagers' missions was its albedo dichotomy: Cassini Regio on the leading hemisphere of Iapetus is dark and its albedo is about 0.04 while the trailing hemisphere has albedo equal to about 0.6 (Hendrix and Hansen, 2007). The albedo dichotomy is widely interpreted as given by deposits of matter originating from the outside of Iapetus (e.g. Clark et al., 1986; Buratti et al., 2002). The albedo dichotomy is sometimes considered as a result of endogenic processes (Smith et al., 1982). New and detailed images of Iapetus's surface, much better than those provided by the Voyagers, were obtained from the Cassini mission during its encounter in December 31, 2004 at minimum altitude 123,400 km. The next close encounter is planned during Orbit 49, on September 10, 2007, at minimum altitude 1600 km.

Large parts of Iapetus were pictured by Cassini with the resolution of the order of 1 km/pixel. The most intriguing

feature discovered by Cassini is a topographic ridge that coincides almost exactly with the geographic equator see Fig. 1. The existence of the ridge raises some problems that are expressed by Porco et al. (2005) as follows: “[...] the presence of a major geological feature of internal origin – the equatorial ridge – bisecting the leading hemisphere might suggest a genetic relationship between the ridge and the darkening of Cassini Regio. Such relationship would require, to begin with, an internal heat source sufficient to drive eruptive volcanism extensive enough to cover an entire hemisphere. Although some degree of volcanism

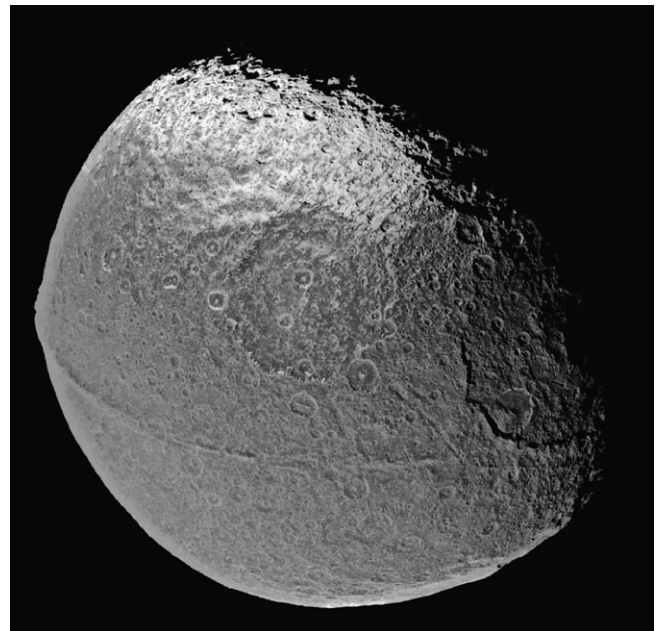


Fig. 1. Photomosaic of Iapetus, PIA06166. Courtesy of NASA.

early in Iapetus's history might have been possible, there is no presently known heat source (such as conventional tidal heating) large enough for such extensive resurfacing of an entire hemisphere postdating the cratering epoch." The goal of our paper is more limited; it presents some hypotheses of origin of the equatorial ridge.

The precise age of equatorial ridge is not known and not crucial for our hypotheses. The above citation from Porco et al. (2005) suggests the formation after Heavy Bombardment Epoch (HBE). Porco et al. (2005) also state: "The ridge is cut by impact craters in some places, which indicated that is not the most youthful surface feature." Let us note also that the ridge is cratered but is not crater-saturated. It is partially broken but not seriously devastated. In fact, identification of the continuity of the ridge is no doubtful, see Fig. 1. It can be supposed, that the ridge was formed during the final stage of HBE or after HBE. Therefore we are looking for the source of energy that could operate in Iapetus interior during the final stage of intensive cratering and that could be able to form the equatorial ridge. We are also considering the plausible mechanisms of its formation.

2. Basic formulae

The large-scale tectonic structures observed on many MIS could be interpreted as resulting from thermal solid-state convection in their interiors below the lithosphere (e.g. Schubert et al., 1986). The lithosphere is an uppermost, cold layer of a satellite that could be treated as an elastic or brittle layer. The material below the lithosphere is warm enough for solid-state creep. The intensity of convection is characterized by a dimensionless Rayleigh number Ra . High Ra corresponds to intensive convection. For the heat sources distributed inside the fluid Ra is defined in Paper I as

$$Ra = \frac{g\alpha c_p \rho^3 d^5 q}{k^2 \eta}, \quad (1)$$

where g , α , c_p , ρ , d , q , k , η , denote surface gravity [m s^{-2}], coefficient of thermal expansion [K^{-1}], specific heat [$\text{J kg}^{-1} \text{K}^{-1}$], density [kg m^{-3}], radius of the satellite [m], average heat production rate per unit mass [W kg^{-1}], coefficient of thermal conductivity [$\text{W m}^{-1} \text{K}^{-1}$], and viscosity coefficient [Pa s], respectively.

Two sources of heat generation could be important: radiogenic heating q_{rad} and tidal heating q_{tidal} , so

$$q = q_{\text{tidal}} + q_{\text{rad}}. \quad (2)$$

The radiogenic heating q_{rad} is related to the silicates content. It is given in Table 1.

The tidal heating is orbit-dependent. It is an important heat source for some MIS (see Paper I). In a synchronously rotating body on an eccentric orbit the global heat production rate [J s^{-1}] is given by the following formula (see e.g. Peale, 2003)

$$\frac{dE_t}{dt} = \frac{42}{19} \pi \rho^2 n^5 d^7 e^2 \frac{1}{\mu Q} = \frac{42}{19} \rho d^3 \chi \frac{1}{\mu Q}, \quad (3)$$

where n is the mean orbital motion [rad s^{-1}], e is the eccentricity of satellite's orbit, Q is the dimensionless dissipation factor (quality factor), μ is the shear modulus [Pa], and

$$\chi = \rho n^5 d^4 e^2 [\text{kg m s}^{-5}]. \quad (4)$$

The shear modulus μ for water ice I is approximately 3.3 GPa (Sotin et al., 1998). Dividing Eq. (3) by satellite mass gives the tidal heat production rate per unit mass q_{tidal}

$$q_{\text{tidal}} = \frac{63\chi}{38\mu Q}. \quad (5)$$

The dissipation factor Q is usually assumed to be equal to 100. However, its true value is virtually unknown. Certainly, the value of Q depends on material properties and temperature.

3. Tectonic regimes: compression versus extension

The solid-state convection is the main factor responsible for origin of internal processes in many planetary bodies. The terrestrial oceanic ridges and subduction zones are results of solid-state convection in the Earth's mantle. In two of three hypotheses of the origin of the Iapetus's ridge we suggest that the ridge is a result of interaction of the lithosphere with convective current. Let us consider this interaction in two different regimes.

- (i) *Interaction with the upward currents (extensional regime)*. In addition to tension, two other factors are involved in the process. They are: increased temperature and penetration of the lithosphere by fluid (cryo-magma). According to the typical geological models (see e.g. Tarbuck and Lutgens, 1988, p. 166) these factors should initially lead to the formation of a ridge with possible magmatic activity above the ascending convective current. Eventually, if the upward current is still active, the ridge evolves to form a tectonic valley (however it is not the case on Iapetus!). The great system of canyons called Ithaca Chasma on Tethys could serve as a good example of such tectonic valley. The Ithaca Chasma extends at least three fourth of the Tethys's circumference. It is a result of an ancient episode of endogenic activity on Tethys. Terrestrial tectonic valleys (e.g. East African Rift) and mid-oceanic ridges are also of extensional origin. A feature of extensional origin is often rectilinear and symmetric with respect to its axis. It is often rather long (extending over the global scale).
- (ii) *Interaction with a downward current (compressional regime)*. The downward current is cold so there is no magmatic activity. Note that volcanic activity along the subduction zones on the Earth is a result of very specific processes that are not expected on

the MISs. The compressional stresses could give rise to thrust faults or folding. It should be noted that strength of the intact rock is not a crucial parameter of the basic theory of thrust faulting (e.g. Watts, 2001, pp. 253–271; Turcotte and Schubert, 2002, pp. 343–350). It is usually assumed that the medium contains many pre-existing joints and faults. Under the tectonic stresses those pre-existing zones of weakness are activated. The effective strength of the rock layer depends mostly on the coefficient of friction and on the weight of the rock layer. Observations of MIS do not indicate many tectonic structures that could be the results of compression. However it is suggested that a few curvilinear ridges on Dione and curvilinear ridge belts on Enceladus resembling fold belts on the Earth could be attributed to compression (Kargel and Pozio, 1996). Compressional origin is also proposed for the lobate scarps on Mercury (Watters and Robinson, 2003).

The stresses resulting from convection depend on many factors and they could reach even 100 MPa (numerical model of convection driven by radiogenic heating in Dione by Forni et al., 1991). This value exceeds the maximum strength of water ice and therefore they are able to fracture the icy lithosphere both in extensional and in compressional regimes. The mean velocity of convection depends on the Rayleigh number. For Ra of the order of 10^5 a typical value of velocity is of the order of 1 mm yr^{-1} (see Paper I). Such velocity of convective flow could lead to displacement on the surface up to 10 km in a period as short as 10^7 years.

Presently available information about Iapetus's ridge is quite limited. However, it seems to be symmetric in respect to its axis and evidently is not curvilinear (it almost follows a great circle). The total length of the ridge is not known but currently documented length is 1400 km therefore one third of the Iapetus circumference (Porco et al., 2005). Moreover, according to Denk et al. (2005): "The ridge itself appears to be located on a broader bulge, extending a few hundred km in southern and northern directions." All these facts fit the hypotheses that the Iapetus's ridge is a structure analogous to the terrestrial ridges frozen in the initial stage of development (compare e.g. Tarbuck and Lutgens, 1988, p. 166; Watts, 2001, p. 286). Therefore we suggest that the ridge is a result of an ancient episode of initial stage of extension. The bulge could be a result of an upward convective current developed below the lithosphere; the ridge itself could be a result of penetration of the lithosphere by cryo-magmas. The similarity of tectonic processes on the Earth and on the icy satellites is not especially surprising. There are other similarities. Giese et al. (2006) state about Ithaca Chasma on Tethys that: "The most striking feature of the Chasma is the upraised flank topography that is elevated up to 6 km above the surroundings. Flank-uplift extends over about 60 km across

strike direction." and "upraised flank topography is typical for rifts on Earth and was also found for rifts on the icy surface of Ganymede." Let us note that Ithaca Chasma extends along great circle like the Iapetus's ridge does. We suggest that the Iapetus's ridge and Ithaca Chasma could be the structures of similar origin. The evolution of the Iapetus's ridge had stopped at an early stage while Ithaca Chasma was developing to more mature stage forming a tectonic valley. Of course, this hypothesis may be revised when more details of the Iapetus's surface will be revealed.

4. Convective patterns

According to McKinnon (1998): "Convection was probable (if not vigorous) in all but the smallest middle size icy satellites early in solar system history." Iapetus is evidently not the smallest one of the MIS; in fact it is one of the largest of MIS. Numerical calculations with temperature-dependent viscosity given by Forni et al. (1991) indicate that convection is possible in Dione. The latter is smaller than Iapetus. Note that the Rayleigh number is proportional to the fifth power of the satellite radius; see Eq. (1). So, if convection is possible in Dione then it will be possible in Iapetus as well.

The patterns of convection driven by tidal heating and of convection driven by exclusively radiogenic heating are generally similar for the range of Rayleigh numbers appropriate for the MISs (Paper I). The patterns are approximately axially symmetric. Two patterns are possible: One-cell pattern and two-cell pattern, Fig. 2. The one-cell pattern leads to bipolar asymmetry (dichotomy between the tectonics of the two hemispheres of the satellite). In the region above an upward flow, extensional surface features and volcanic activity are expected while in the region above a downward flow compressional surface features are expected. A more complicated case is the two-cell convection: there are two regions of compressional tectonics separated by a belt of extensional tectonics. In that case the most important tectonic feature that could accompany convection is a belt of extensional tectonics ridge extending

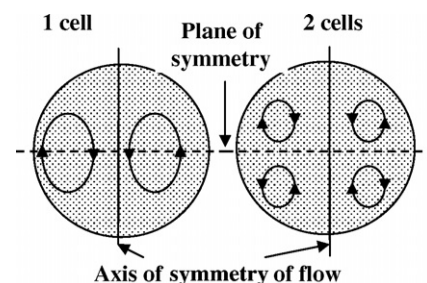


Fig. 2. Two basic patterns of convection characteristic for low and moderate Rayleigh numbers: one-cell pattern and two-cell pattern. The convective patterns have a plane of symmetry ('equatorial region') perpendicular to the axis of symmetry. In reality the symmetry of the convective pattern is not perfect.

over the circumference of the satellite. So, the two-cell pattern seems to be more probably responsible for the origin of the Iapetus's ridge than one-cell pattern. The ridge is formed above the region of extensional tectonics. Note that the surface manifestation of this convection pattern could be masked by the thick, resistant lithosphere or could be damaged by impacts.

Radioactivity and other spherically symmetric heat sources do not prefer any orientation of the convective pattern. Note that rotation of the satellite has negligible influence on the orientation of the convective pattern because the velocity of solid-state convection is very low (of the order of 1 mm per year). However, tidal heating is not spherically symmetric and it leads to convective pattern strongly oriented with respect to the planet–satellite line. Let us consider the two-cell pattern. If the tidal heating is a result of eccentricity of the satellite orbit then the region of extensional tectonics lies in the plane perpendicular to the planet–satellite line. The axis of symmetry of convection should be oriented towards the planet. The ridge would follow the great circle formed by the leading and trailing meridians. If the tidal heating is a result of inclination of the orbit then the region of extensional tectonics lies in the plane of the planet–satellite line and the satellite's axis of rotation. The present orientation of the Iapetus's ridge (in its equatorial plane) does not fit any of the orientations mentioned above. However, this fact could be explained as a result of mass redistribution inside the satellite. The mass redistribution resulting from convection could lead to instability and to reorientation of the satellite. Note that the reorientation is not a cataclysmic event. Therefore, we suggest that after a relatively short period of internal activity Iapetus had changed orientation with respect to the planet–satellite line reaching the present position.

Let us also note that the tidal bulge resulting from a tidal interaction with the planet is usually more massive than most of the surface features (e.g. the ridge) or non-hydrostatic density non-homogeneities (e.g. resulting from convection) in the satellite interior. However, the bulge does not prefer any satellite orientation and does not preclude changes of orientation. If the density redistribution resulting from origin of a ridge and/or convection would prefer change satellite orientation then the tidal bulge will move according to changing orientation of a satellite.

5. Internal heat sources

There are a few potential heat sources in the MIS. They are: primary heat of accretion, self-compaction energy, radiogenic heat, energy of differentiation, energy of phase changes, and tidal heating. All these sources were not intensive enough to give rise to the long lasting endogenic activity on another Saturnian satellite Rhea (see also tectonic activity parameter in Table 1). It is a little larger and a little more massive than Iapetus. Note that the efficiency of the primary heating due to accretion, self-com-

paction energy, energy of differentiation, energy of phase changes is higher for more massive bodies. The radiogenic heating rate is also higher in Rhea than in Iapetus (see Table 1). Therefore, it is unlikely that the mentioned sources, (independent of the satellite orbit), could be responsible for the endogenic activity on Iapetus after the HBE.

Aluminum Al^{26} , could be a powerful source of energy (e.g. Prialnik and Bar-Nun, 1990). The assumption that during formation of the Solar System bodies the Al^{26} was present is very convenient for many hypotheses requiring a substantial heat input. However, the Iapetus's ridge seems to be a structure a few hundreds Myr younger and therefore we cannot treat the ridge as a direct consequence of Al^{26} heating. Of course we do not reject the possibility that Al^{26} decay could be important for thermal history of Iapetus (as well as of other MIS). In this paper we do not discuss scenario of Iapetus evolution supported by presence of Al^{26} .

Let us consider the tidal heating, for Rhea and Iapetus. Using the formula (5) we consider tidal heating for synchronous rotation. Taking the present-day values of the known parameters (ρ, n, d, e) and assuming that the values of material parameters (Q, μ) are equal for both satellites one can calculate the ratio of the tidal heating rates for both bodies. The results presented in Table 1 indicate that the tidal heating rate for Rhea is three orders of magnitude higher than for Iapetus and it is of the same order for Tethys. Therefore Iapetus on its present-day orbit seems to be a worse candidate for a body with tidally driven convection.

6. Evolutionary scenarios

The discussion presented above indicates that Iapetus should not exhibit the large scale traces of the internal global flow. The observation shows the opposite. So, for the interpretation of the Iapetus evolution the special circumstances explaining the effectiveness of the heat sources and/or rheology are necessary. We are looking for a hypothesis explaining the origin of Iapetus's ridge that allows Rhea to be dead. Moreover we would like to avoid of introduction of any *ad hoc* assumptions. To explain the existence of the mysterious ridge on Iapetus we propose three different scenarios. The first scenario is based on the possible spin–orbit resonance, the second scenario is based on the assumption that the viscosity of Iapetus's interior is lower than in the interior of Hera, and the third one is just adaptation of the McKinnon (1985) impact scenario. It seems that all of them are reasonable although the third scenario appears less likely.

6.1. Spin–orbit resonance scenario

We propose the following sequence of events:

1. Accretion of Iapetus on an elliptical orbit. The density distribution and/or the shape of Iapetus are not exactly spherically symmetric resulting in asymmetry of Iapetus's gravitational field. See notes concerning the shape of Iapetus presented in Section 1 and (v) in Section 7.
2. After accretion a series of consecutive spin–orbit resonances appears. Each of the resonances locks the rotation rate constant for some time interval. (Compare the case of spin–orbit resonance 2:3 of Mercury.) Consecutive resonances extend the time of despinning. So, the state of synchronous rotation is still not achieved. The fast spinning results in intensive tidal heating. Interior temperature is increasing causing the onset of convection. The convection forms the ridge. Convective heat transfer led to intensive cooling of the globe and eventually stopped convection. It could happen at that time when Iapetus rotation period was 17 h. A rough estimate of tidal heating rate during resonance could be done in the following way. The rotational energy E of the satellite is

$$E = \frac{1}{2} I \omega^2. \quad (6)$$

Here I is the moment of inertia of the satellite, and ω is the angular velocity of rotation. Note that $\omega = 2\pi/P_{\text{rot}}$ where P_{rot} is the rotational period of the satellite. Differentiating Eq. (6) with respect to time we get the energy dissipation rate

$$\frac{dE}{dt} = I \omega \frac{d\omega}{dt}. \quad (7)$$

The gravity of Saturn leads to deformation of the satellite in the form of a tidal bulge. Due to viscous properties of the satellite's interior the bulge is not situated exactly on the planet–satellite line but it is shifted by a small angle δ (phase shift). The moment of force exerted by the planet on the satellite's bulge is given by the following formula (see e.g. Hubbard, 1984, p. 102)

$$L = \frac{GM^2 d^5 \delta}{r^6}, \quad (8)$$

where $G = 6.672 \cdot 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$ and $M = 5.686 \times 10^{26} \text{ kg}$ is the mass of Saturn. Substitution of $L = I(d\omega/dt)$ into Eq. (7) divided by m gives the heat production rate per unit mass of the satellite

$$\frac{1}{m} \frac{dE}{dt} = \frac{GM^2 d^5 \omega \delta}{mr^6}. \quad (9)$$

The latter formula needs some comments. We neglect tidal deformations of the planet. It is justified because we are looking for rough approximation only. Note that the tidal deformations of the planet are important for evolution of the satellite's orbit but its effect on the tidal deformation of the satellite is moderate. Moreover we assume that most of the rotational energy of the satellite is dissipated in the form of heating of the satellite's interior. Formula (9) has its physical meaning independently of

the behavior of rotation. It means that Eq. (9) is valid for the case of $\omega = \text{const}$ (as it is in a resonance state) as well as in the case when ω continuously decreases due to tidally driven spin-down. Fig. 3 and Table 2 present the heating rates for different angular velocity ω and different distance r from the planet. The phase shift δ is assumed to be 0.01 rad. The table could be applied for the spin orbit resonance. For example; for angular velocity $\omega = 10^{-5} \text{ rad s}^{-1}$ and a distance $r = 2.5 \times 10^6 \text{ km}$ the rate of heat generation is $10^{-12} \text{ W kg}^{-1}$. This value denotes a rate of heating per unit of mass for the whole period of the considered resonance. For present distance the heating rate is comparable or substantially lower than radiogenic heating depending on angular velocity ω (see Table 1). Note however that the phase shift $\delta = 0.01 \text{ rad}$ is rather a moderate value; depending on rheological properties δ could be substantially higher leading to higher rate of tidal dissipation. The described mechanism includes the heat released in the process of despinning as well as the energy provided by the resonance. The total specific energy available from simple despinning is $0.2 d^2 \omega_{\text{initial}}^2$ for a uniform body. Our proposed spin–orbit resonance mechanism has two important advantages comparing to the simple despinning: (i) the total dissipated energy could be higher than energy of despinning because in addition to energy of despinning there are some energy supplied by the resonance, (ii) the most of energy could be dissipated during the final stage of HBE when Iapetus's ridge was probably formed.

3. Breaking the resonance. This does not require any dramatic events. It could be a result of an orbit evolution (e.g. decreasing of eccentricity) or of an interior evolution (e.g. cooling of the interior and consequently changing its elastic properties). Both processes are expected. The breaking of the resonance could be also a result of impact. Of course, all factors could combine. After breaking the resonance the energy inflow due to the resonance is stopped. Iapetus spins down due to tidal interaction with Saturn into a lower degree resonance. At some resonance the tidal heating is so low that convection stops. It leaves the unstable orientation of the ridge with respect to the Iapetus–Saturn line. For orientation of the axes of symmetry of convective flow versus Iapetus–Saturn line, see Paper I.
- 4/5 Iapetus reaches its present state of synchronous rotation (1:1). The energy dissipation due to tidal deformation becomes negligible.
- 4/5 Reorientation of Iapetus into stable present day position: The Iapetus–Saturn line passes through the plane of the equatorial ridge.

6.2. Convection in low-viscosity-interior scenario

The above considerations do not include the possibility of different rheological properties of Iapetus and Rhea. However, such a possibility exists. The conventional scenario of

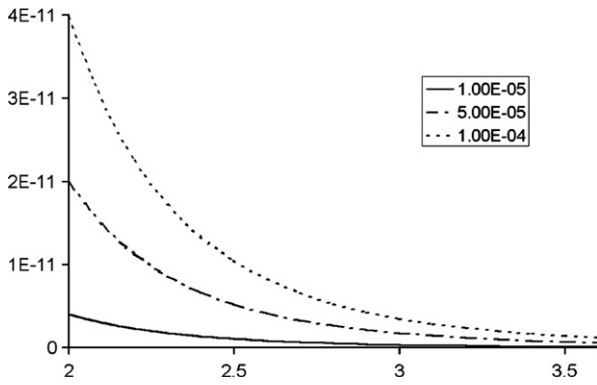


Fig. 3. The heat production rate, Eq. (9), in W kg^{-1} (vertical axis) in Iapetus during hypothetical spin–orbit resonance for different values of semi-major axes r (horizontal axis) and for different angular velocities ω : $10^{-5} \text{ rad s}^{-1}$ (solid line), $5 \times 10^{-5} \text{ rad s}^{-1}$ (dashed-dotted line), and $10^{-4} \text{ rad s}^{-1}$ (dotted line). The phase shift δ is assumed to be 0.01 rad. See also Table 2.

formation of the regular icy satellites from circumplanetary nebula takes into account the decreasing temperature of the nebula with increasing distance from the planet. So, one can expect that more volatile substances were caught in satellites that accreted farther from the planet. Consequently, Iapetus could be enriched with ammonia compared to Rhea (perhaps with other, non-water volatiles as well). Note, that ammonia was identified by spectroscopic methods on the surfaces of two cold bodies: Miranda and Charon (Brown and Calvin, 2000; Roush, 2001). Hypothesis of the presence of ammonia in the interiors of icy satellites is widely discussed. Even small admixture of ammonia could change significantly the rheology of the satellite interior. For some experimental data concerning rheology of the water–ammonia systems see Leliwa-Kopystyński and Kossacki (1995). So, the considered scenario postulates lower viscosity of Iapetus's interior compared to other MISs. Lower viscosity implies higher value of the Rayleigh number (Eq. (1)). Consequently, convection can be expected even without any additional heat sources, i.e., only primordial heat and radiogenic heat would be sufficient. The last stage of this scenario is the reorientation of Iapetus into stable, present-day position mentioned in Scenario 6.1.

The problem of ammonia (as well as other possible volatiles) could be discussed in the frame of general rheological properties. Note that ammonia enrichment changes viscosity as well as the quality factor Q and consequently

δ (according to Hubbard (1984) for small phase shift $\delta \approx 1/Q$). The change of Q leads to changes of the rate of energy dissipation (Eq. (9)). Presently, there are not observational data for any of MIS that could give a reliable value of the viscosity η or quality factor Q . The experimental and theoretical papers concerning the viscosity of the water ice or the water ice enriched by other volatiles indicate that the viscosity depends on many factors: temperature, grain size, volatiles concentrations, and rate of deformations. According to Barr and Pappalardo (2005) the effective viscosity of water ice I is given by

$$\eta(T, x) = \left(\frac{x^p}{2A} \right)^{1/i} \varepsilon_{\text{II}}^{(1-j)/j} \exp \left(\frac{E^*}{jRT} \right). \quad (10)$$

Here x is the ice grain size, p is the grain size exponent, A is the pre-exponential parameter, ε_{II} is the second invariant of the strain rate tensor, j is the stress exponent, E^* is the activation energy, R is the gas constant, and T is the temperature. Parameters in Eq. (10) depend on the deformation mechanism. Barr and Pappalardo (2005) use: (i) for the volume diffusion $A = 1.2 \times 10^{-10}$, $j = 1$, $p = 2$, $E^* = 59,400 \text{ J mol}^{-1}$. (ii) For grain boundary sliding $A = 6.2 \times 10^{-14}$, $j = 1.8$, $p = 1.4$, $E^* = 60,000 \text{ J mol}^{-1}$. In our considerations we assume $\varepsilon_{\text{II}} = 10^{-16} \text{ s}^{-1}$ that corresponds to convection velocity 1 mm yr^{-1} , approximately (see Paper I). For the assumed ε_{II} the other deformation mechanisms seem to be less important.

Let us now consider Rayleigh number Ra . The convection that could give origin to the Iapetus's ridge cannot be too weak. It should be intensive enough to result in some surface features. Let us consider the Rayleigh number during the ridge formation. It evidently must exceed substantially 10^4 ($Ra \approx 10^4$ is a critical value for convection driven by spherically symmetric sources; Paper I). On the other hand, the convection could not be intensive. Note that evolution of the ridge had stopped at the early stage of evolution (see Section 3). Therefore the reasonable value of Ra during the ridge formation was of the order of 10^5 . Using $Ra = 10^5$ in Eq. (1), and adopting standard values of the parameters α , c_p , and k (see Paper I), we can conclude that during the formation of the Iapetus's ridge the effective average viscosity of the satellite interior was in the range of 10^{19} – 10^{20} Pa s . Note that we take into account a higher rate of heat production than the present rate (radiogenic heating q was more intensive). That range of viscosity and

Table 2

The heat production rate, Eq. (9) in W kg^{-1} , in Iapetus for different values of semi-major axes r and for different angular velocities ω

Orbital radius r in 10^6 km	Angular velocity ω in rad s^{-1} of the satellite and the corresponding rotation period P_{rot} in days			
	$\omega = 5.0\text{E-}06$, $P_{\text{rot}} = 14.54$	$\omega = 1.0\text{E-}05$, $P_{\text{rot}} = 7.272$	$\omega = 5.0\text{E-}05$, $P_{\text{rot}} = 1.454$	$\omega = 1.0\text{E-}04$, $P_{\text{rot}} = 0.7272$
2.0	1.91E-12	3.83E-12	1.91E-11	3.83E-11
2.5	5.02E-13	1.00E-12	5.02E-12	1.00E-11
3.0	1.68E-13	3.36E-13	1.68E-12	3.36E-12
3.5	6.66E-14	1.33E-13	6.66E-13	1.33E-12
3.561 the present-day value	6.01E-14	1.20E-13	6.01E-13	1.20E-12

The phase shift δ is assumed to be 0.01 rad. See also Fig. 3.

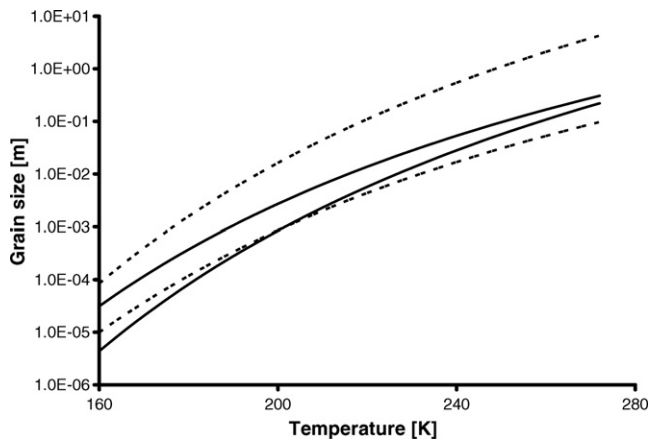


Fig. 4. The grain size versus temperature, according to formula (10) for assumed range of viscosity 10^{19} – 10^{20} Pa s. Upper dotted line is for the grain boundary sliding and for $\eta = 10^{20}$ Pa s. Lower solid line is for the grain boundary sliding and for $\eta = 10^{19}$ Pa s. Upper solid line is for the volume diffusion creep and for $\eta = 10^{20}$ Pa s. Lower dotted line is for the volume diffusion creep and for $\eta = 10^{19}$ Pa s.

Eq. (10) allow us to find some limits for the range of the grain size x versus temperature T , see Fig. 4. The temperature range corresponds, approximately, to that given by Schubert et al. (1986) for thermal models of evolution of Iapetus. The range of the grain size (from 0.01 mm to 10 cm) is realistic. Concluding these considerations we would like to state that the present knowledge about rheology of MISs, does not exclude the possibility of convection. The ammonia content makes the possibility even higher.

6.3. Impact-generated flow scenario

The impact-generated scenario is based on hypothesis of McKinnon (1985) and McKinnon and Benner (1989). They looked for the correlation between an impact event that produced Odysseus crater on Tethys and the formation of the Ithaca Chasma. They suggest that viscous rebound of Odysseus crater could cause the observed fracturing (Ithaca Chasma) by inducing viscous drag at the base of the lithosphere. Recently the mechanism relating the formation of Ithaca Chasma to the Odysseus impact are challenged by the fact that another basin on Tethys (almost as large as Odysseus) did not lead to formation of the structure similar to Ithaca Chasma (Giese et al., 2006). The main difficulty in applying the McKinnon's hypothesis to Iapetus is the lack of a known crater centered on one of the poles of Iapetus. The geographic poles of Iapetus are also the poles of its ridge. Note that the ridge follows a large part of the great circle. The last stage of this scenario is the reorientation of Iapetus into stable, present day position mentioned in scenario 1 (Section 6.1).

7. Conclusions

Let us summarize the results and the proposed scenarios.

- (i) The Iapetus's ridge running along the great circle seems to be a feature of extensional origin. The ridge can be simply interpreted as a result of two-cell convection.
- (ii) The spin–orbit resonance could explain a substantial role of tidal heating in Iapetus. A significant role of this mode of heating is important for our hypothesis concerning origin of the Iapetus's ridge since 2-cell pattern is preferred for the tidal heating. See the scenario 6.1 or combination of scenarios 6.1 and 6.2.
- (iii) To realize the impact generating flow scenario 6.3 a large impact on one of the poles is necessary. Unfortunately the scar of such impact is not identified.
- (iv) The ammonia content in Iapetus is expected. The presence of ammonia supports plausibility of the scenario 6.2. A combination of the scenarios 6.1 and 6.2 seems to be the most plausible.
- (v) The large-scale non-homogeneities of density indicated by present orientation of Iapetus ellipsoid is consistent with our assumption about initial asymmetric shape or/and density distribution in Iapetus.

Acknowledgments

This work was supported by the Polish Ministry of Education and Science Grant No. N203 007 32/0576. We are grateful to the anonymous reviewers for their very detailed discussion.

References

- Barr, A.C., Pappalardo, R.T. Onset of convection in the icy Galilean satellites: influence of rheology. *J. Geophys. Res.* 110, E12005, 1–14, doi:10.1029/2004JE002371, 2005.
- Brown, M.E., Calvin, W.M. Evidence for crystalline water and ammonia ices on Pluto's satellite Charon. *Science* 287, 107–109, 2000.
- Buratti, B.J., Hicks, M.D., Tryka, K.A., Sittling, M.S., Newburn, R.L. High-resolution 0.33–0.92 μ m spectra of Iapetus, Hyperion, Phoebe, Rhea, Dione, and D-type asteroids: how are they related? *Icarus*, 155, 375–381, 2002.
- Clark, R.N., Fenale, F.P., Gaffey, M.J. Surface composition of natural satellites, in: Burns, J.A., Matthews, M.S. (Eds.), *Satellites*. University Arizona Press, Tucson, pp. 437–491, 1986.
- Czechowski, L. Planetology and classification of the solar system bodies. *Adv. Space Res.*, 38, doi:10.1016/j.asr.2006.09.004, 2054–2059, 2006.
- Czechowski, L., Leliwa-Kopystyński, J. Solid state convection in the icy satellites: discussion of its possibility. *Adv. Space Res.* 29 (5), 751–756, 2002a.
- Czechowski, L., Leliwa-Kopystyński, J. Solid state convection in the icy satellites: numerical results. *Adv. Space Res.* 29 (5), 757–762, 2002b.
- Czechowski, L., Leliwa-Kopystyński, J. Tidal heating and convection in the medium sized icy satellites. *Celestial Mech. Dyn. Astronomy* 87, 157–169, 2003.
- Czechowski, L., Leliwa-Kopystyński, J. Convection driven by tidal and radiogenic heating in medium size icy satellites (Paper I). *Planet. Space Sci.* 53, 749–769, 2005.
- Denk, T. 12 colleagues, and the Cassini ISS Team. The first six months of Iapetus observations by the Cassini ISS camera. *LPSC XXXVI*, 2262.pdf, 2005a.
- Durham, W.B., Kirby, S.H., Stern, L.A. Flow of ices in the ammonia–water system. *J. Geophys. Res.* 98 (B10), 17667–17682, 1993.

- Forni, O., Coradini, A., Federico, C. Convection and lithospheric strength in Dione, an icy satellite of Saturn. *Icarus* 94, 232–245, 1991.
- Giese, B., Wagner, R., Neukum, G., Helfenstein, P., Porco, C.C. Topographic features of Ithaca Chasma, Tethys. *LPSC XXXVII*, 1749.pdf, 2006.
- Hendrix, A.R., Hansen, C.J. Iapetus, Hyperion and Phoebe: Comparison from Cassini UVIS. *LPSC XXXVIII*, 1916.pdf, 2007.
- Hubbard, W.B. *Planetary Interiors*. Van Nostrand Reinhold Company, New York, 1984.
- Hunten, D.N., Tomasko, M.G., Flasar, F.M., Samuelson, R.E., Strobel, D.F., Stevenson, D.J. Titan, in: Gehrels, T., Matthews, M.S. (Eds.), *Saturn*. University of Arizona Press, Tucson, pp. 671–759, 1984.
- Kargel, J.S., Pozio, S. The volcanic and tectonic history of Enceladus. *Icarus* 119, 385–404, 1996.
- Leliwa-Kopystyński, J., Makkonen, L., Erikoinen, O., Kossacki, K.J. Kinetics of pressure-induced effects in water/ice granular mixtures and application to the physics of the icy satellites. *Planet. Space Sci.* 42, 545–555, 1994.
- Leliwa-Kopystyński, J., Kossacki, K.J. Kinetics of compaction of granular ices H_2O , CO_2 , and $(\text{NH}_3)_x(\text{H}_2\text{O})_{(1-x)}$ at pressures 2–20 MPa and in temperatures 100–270 K. Application to the physics of the icy satellites. *J. Planet. Space Sci.* 43 (7), 851–861, 1995.
- Leliwa-Kopystyński, J., Kossacki, K.J. Evolution of porosity in small icy bodies. *Planet. Space Sci.* 48, 727–745, 2000.
- McKinnon, W.B. Origin of Ithaca Chasma. *Bull. Am. Astron. Soc.* 17, 922, 1985.
- McKinnon, W.B., Benner, L.A.M. Origin of Ithaca Chasma, Tethys II: the importance of the lithosphere. *Abstracts of the LPSC XX*, 679, 1989.
- McKinnon, W.B. Geodynamics of icy satellites, in: *Solar System Ices*. Kluwer Academic Publishers, pp. 525–550, 1998.
- Peale, S.J. Tidally induced volcanism. *Celestial Mech. Dyn. Astronomy* 87, 129–155, 2003.
- Porco, C.C. and 34 co-workers Cassini imaging science: initial results on Phoebe and Iapetus. *Science* 307, 1237–1242, 2005.
- Prentice, A.J.R. Saturn's icy moons: a model for their origin and bulk chemical composition. *LPSC XXXVI*, 2378.pdf, 2005.
- Prialnik, D., Bar-Nun, A. Heating and melting of small icy satellites by the decay of Al^{26} . *Astrophys. J.* 355, 281–296, 1990.
- Roush, T.L. Physical state of ices in the outer solar system. *J. Geophys. Res.* 106 (E12), 33315–33323, 2001.
- Schubert, G., Spohn, T., Reynolds, R.T. Thermal histories, compositions and internal structures of the moons of the Solar System, in: Burns, J.A., Matthews, M.S. (Eds.), *Satellites*. University Arizona Press, Tucson, pp. 224–292, 1986.
- Smith, B.A. and 26 co-workers A new look at the Saturn system: The Voyager 2 images. *Science* 215, 504–537, 1982.
- Sotin, C., Grasset, O., Beauchesne, S. Thermodynamic properties of high pressure ices: Implications for the dynamic and internal structure of large icy satellites, in: Schmitt, B., de Bergh, C., Festou, M. (Eds.), *Solar System Ices*. Kluwer Acad. Publ., pp. 119–138, 1998.
- Tarback, E.J., Lutgens, F.K. *Earth Science*. Merrill Pub. Co. Columbus, Ohio, USA, pp. XII+612, 1988.
- Thomas, P.C., Veverka, J., Helfenstein, P., Porco, C., Burns, J., Denk, T., Turtle, E., Jacobson, R.A. and the ISS team. Shapes of the Saturnian icy satellites. *LPSC XXXVII*, 1639.pdf, 2006.
- Turcotte, D.L., Schubert, G. *Geodynamics: Applications of Continuum Physics to Geological Problems*. Cambridge University Press, Cambridge UK, pp. IX+456, 2002.
- Watters, T.R., Robinson, M.S. The spatial and azimuthal distribution of lobate scarps and high-relief ridges on Mercury. *LPSC XXXIV*, 1927.pdf, 2003.
- Watts, A.B. *Isostasy and Flexure of the Lithosphere*. Cambridge University Press, Cambridge UK, pp. XII+458, 2001.